Progress Report: Implementation of Transformers on Graph Data (METR-LA Dataset)

Course: MA-INF 4325 - Lab Data Science in Practice: Spatio-Temporal Data Analytics

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Overview

I have been working on implementing a Graph Transformer model to analyze and predict traffic speed data from the METR-LA dataset, which consists of spatio-temporal traffic measurements from 207 sensors in Los Angeles. The goal is to leverage the graph structure of the road network and temporal dependencies using a Transformer-based architecture to improve traffic forecasting accuracy.

Progress to Date

1. **Data Preparation and Loading**:

   - Successfully loaded and preprocessed the METR-LA dataset, including the adjacency matrix (shape [207, 207]) and node values (shape [34272, 207, 2]), extracting the speed feature and transposing it to [207, 34272] for normalization.
   - Implemented normalization using MinMaxScaler to scale the data between 0 and 1, ensuring consistency across the dataset.
   - Converted the adjacency matrix into edge_index and edge_weight tensors for use with PyTorch Geometric, utilizing GPU acceleration where available.

2. **Model Development**:

   - Designed a GraphTransformer class using PyTorch Geometric's TransformerConv layer, incorporating positional encoding via AddRandomWalkPE to capture spatial relationships.
   - The model architecture includes an embedding layer, multiple transformer convolution layers (4 layers with 4 heads each), batch normalization, and a final MLP for output prediction.

- Configured the model with 12 input channels (timesteps), 64 hidden channels, and 12 output channels (prediction horizon), trained for 20 epochs with the AdamW optimizer and MSE loss.

3. **Training and Evaluation**:

   - Split the dataset into training (70%), validation (20%), and test (10%) sets, using a batch size of 32.
   - Achieved a training loss of 0.0209 and a validation loss of 0.2131 after 20 epochs, indicating reasonable convergence.
   - Evaluated the model on the test set, obtaining a Mean Absolute Error (MAE) of 0.4090, Root Mean Squared Error (RMSE) of 0.4387, and Mean Absolute Percentage Error (MAPE) of 358045710.7407%. The high MAPE suggests potential issues with the metric calculation or model scaling, which I am investigating.

4. **Visualization**:

   - Implemented a visualize_predictions function to plot predicted vs. actual traffic speeds for the first three sensors over a 12-timestep horizon (5-minute intervals).
   - Successfully generated and saved plots as prediction_plot.png in the plots directory, with the plots showing reasonable alignment between predicted and actual values, though some discrepancies are evident (e.g., a notable drop in actual values around timestep 6).

Challenges Encountered

- **Shape Mismatch Error**: Initially encountered a ValueError: non-broadcastable output operand with shape (12,1) doesn't match the broadcast shape (12,207) during inverse transformation in the visualization function. This was resolved by correctly transposing the prediction and target tensors to match the scaler's expected input shape.
- **High MAPE**: The extremely high MAPE value suggests a potential issue with the compute_metrics function, possibly due to division by very small or zero target values. I plan to debug this by adding checks for zero values and refining the metric calculation.
- **Plot Visibility**: Initially, plots were not displayed in the notebook; adding plt.show() resolved this in the Colab environment, and files are now accessible in the plots directory.

Next Steps

- **Debug and Refine Metrics**: Investigate the high MAPE value by reviewing the compute_metrics function, adding safeguards for small target values, and potentially normalizing the MAPE calculation.

- **Model Optimization**: Experiment with hyperparameter tuning (e.g., learning rate, number of layers, heads) and additional regularization to improve validation loss and test metrics.
- **Enhanced Visualization**: Extend the plotting function to visualize multiple batches or include confidence intervals if feasible, and ensure plots are consistently displayed or saved for review.
- **Report and Documentation**: Prepare a detailed report with updated results and visualizations, including a comparison with baseline models (e.g., simple temporal models or GCNs), to be submitted by the next milestone.

Conclusion

I have made significant progress in implementing the Graph Transformer model for the METR-LA dataset, overcoming initial data and visualization challenges. The current model shows promise, but further refinement is needed to address the high MAPE and optimize performance. I am on track to meet the project goals and look forward to your feedback to guide my next steps.

Attachments

prediction_plot.png (included in the plots directory of the submitted notebook).